

Neural Machine Translation

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Outline

NMT History

Sequence to Sequence Translation

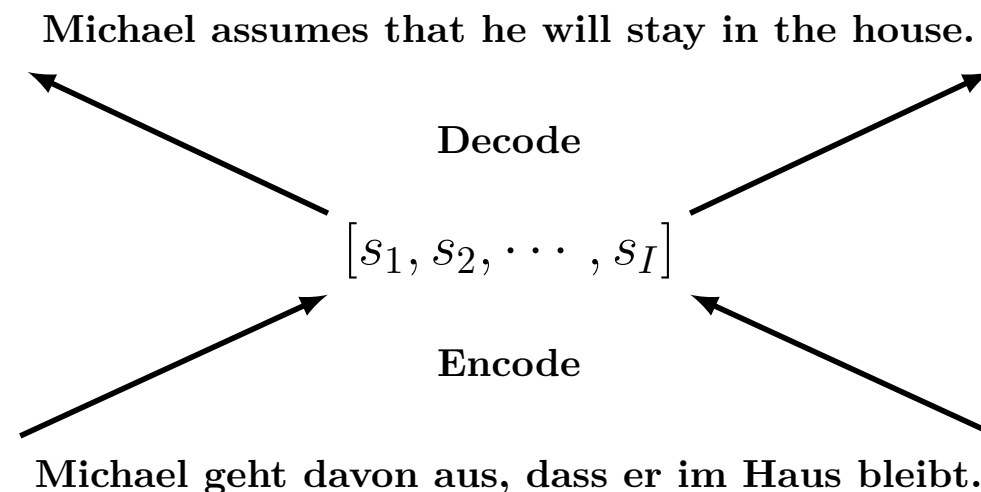
Comparison

NMT History

- 2014 • Sequence to sequence translation
[Sutskever & Vinyals⁺ 14,
Cho & van Merriënboer⁺ 14b]
- 2015 • Attention-based neural machine translation
[Bahdanau & Cho⁺ 15, Luong & Pham⁺ 15a]
- 2016 • Subword-NMT [Sennrich & Haddow⁺ 16b]
• Multi-lingual NMT [Firat & Cho⁺ 16]
• WMT 2016
• Google NMT [Wu & Schuster⁺ 16]
• Fully char-level NMT [Lee & Cho⁺ 16]
- 2017 • ...

Sequence to Sequence Translation

- ▶ Basic principle behind neural machine translation
- ▶ Translation from one language to another can be done by **a single neural network**
- ▶ The input and output are both variable-length sequences
- ▶ An encoder reads the source sentence, encodes it into a vector
- ▶ Decoder uses this vector to predict the target words



Comparison to Best Systems

	WMT2015 De→En newstest2014		WMT2016 En→Ro newstest2016		BOLT Zh→En test1	
	BLEU %	TER %	BLEU %	TER %	BLEU %	TER %
Model A	28.1	53.2	24.5	59.3	17.9	67.7
Model B	24.0	57.2	26.7	56.6	17.9	67.2
Model C	28.2	53.0	25.3	58.4	17.6	68.9
Model D	29.2	52.9	27.1	55.4	21.7	63.7

- ▶ **De→En and Zh→En:**
 - ▷ **Subwords and LSTM**

Thank you for your attention

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Results: Cool Method with Overlay (IWSLT2013)

IWSLT De-En	dev		test		eval11		alignment-test	
Model	BLEU %	TER %	BLEU %	TER %	BLEU %	TER %	AER%	SAER%
Attention-Based	30.5	48.7	29.3	50.6	33.9	46.6	41.8	66.3
+ method	31.5	47.2	30.3	49.0	34.3	44.3	35.4	44.2

- ▶ Improves translation by an average of 0.8 BLEU on IWSLT2013
- ▶ Great improvement in AER and SAER

Results: Cool Method with Overlay (WMT2016)

WMT En-Ro	newsdev2016/1		newsdev2016/2		newstest2016	
Model	BLEU %	TER %	BLEU %	TER %	BLEU %	TER %
Attention-Based	19.8	62.0	21.3	58.1	20.3	60.4
+GA	21.0	61.1	23.6	56.4	21.8	59.4
+method + conv ($D = 10, M = 1$)	21.4	60.1	24.7	55.4	22.3	58.7

- ▶ Improves translation by an average of 0.8 BLEU on IWSLT2013
- ▶ Great improvement in AER and SAER
- ▶ Improves translation by an average of 1.7 BLEU on WMT2016
 - ▷ Adding convolutional feedback gives an additional 0.6 BLEU on average

Reference

-  **D. Bahdanau, K. Cho, Y. Bengio.**
Neural machine translation by jointly learning to align and translate.
In Proceedings of the International Conference on Learning Representations (ICLR), San Diego, CA, 2015.
-  **Y. Bengio, R. Ducharme, P. Vincent, C. Jauvin.**
A neural probabilistic language model.
journal of machine learning research, Vol. 3, No. Feb, pp. 1137–1155, 2003.
-  **M. Castana, E. Vidal, F. Casacuberta.**
Inference of stochastic regular languages through simple recurrent networks.
In Grammatical Inference: Theory, Applications and Alternatives, IEE Colloquium on, pp. 16–1. IET, 1993.
-  **W. Chen, E. Matusov, S. Khadivi, J. Peter.**
Guided alignment training for topic-aware neural machine translation.
CoRR, Vol. abs/1607.01628, 2016.

-  **W. Chen, E. Matusov, S. Khadivi, J.-T. Peter.**
Guided alignment training for topic-aware neural machine translation.
Austion, Texas, October 2016. Association for Machine Translation in the Americas.
-  **K. Cho, B. van Merrienboer, D. Bahdanau, Y. Bengio.**
On the properties of neural machine translation: Encoder-decoder approaches.
In Proceedings of SSST-8, Eighth Workshop on Syntax, Semantics and Structure in Statistical Translation, pp. 103—111, Doha, Qatar, October 2014.
-  **K. Cho, B. van Merrienboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, Y. Bengio.**
Learning phrase representations using rnn encoder–decoder for statistical machine translation.
In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 1724–1734, Doha, Qatar, October 2014. Association for Computational Linguistics.



K. Cho.

Natural language understanding with distributed representation.
Technical report, New York University, 2015.
Lecture Note for DS-GA 3001.



J. K. Chorowski, D. Bahdanau, D. Serdyuk, K. Cho, Y. Bengio.

Attention-based models for speech recognition.
In *Advances in Neural Information Processing Systems*, pp. 577–585,
2015.



J. K. Chorowski, D. Bahdanau, D. Serdyuk, K. Cho, Y. Bengio.

Attention-based models for speech recognition.
In C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, R. Garnett, editors,
Advances in Neural Information Processing Systems 28, pp. 577–585.
Curran Associates, Inc., 2015.

- 📄 **J. Chung, K. Cho, Y. Bengio.**
A character-level decoder without explicit segmentation for neural machine translation.
Proceedings of Association for Computational Linguistics (ACL), Vol., pp. 1693–1703, 2016.
- 📄 **J. Chung, K. Cho, Y. Bengio.**
A character-level decoder without explicit segmentation for neural machine translation.
In Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics, ACL 2016, August 7-12, 2016, Berlin, Germany, Volume 1: Long Papers, 2016.
- 📄 **T. Cohn, C. D. V. Hoang, E. Vymolova, K. Yao, C. Dyer, G. Haffari.**
Incorporating structural alignment biases into an attentional neural translation model.
CoRR, Vol. abs/1601.01085, 2016.

- 📄 **O. Firat, K. Cho, B. Sankaran, F. T. Y. Vural, Y. Bengio.**
Multi-way, multilingual neural machine translation.
Computer Speech & Language, Vol., 2016.
- 📄 **Ç. Gülçehre, O. Firat, K. Xu, K. Cho, L. Barrault, H. Lin, F. Bougares, H. Schwenk, Y. Bengio.**
On using monolingual corpora in neural machine translation.
CoRR, Vol. abs/1503.03535, 2015.
- 📄 **S. Jean, K. Cho, R. Memisevic, Y. Bengio.**
On using very large target vocabulary for neural machine translation.
In Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), pp. 1–10, Beijing, China, July 2015. Association for Computational Linguistics.
- 📄 **J. Lee, K. Cho, T. Hofmann.**
Fully character-level neural machine translation without explicit segmentation.
CoRR, Vol. abs/1610.03017, 2016.

- 📄 **M.-T. Luong, H. Pham, C. D. Manning.**
Effective approaches to attention-based neural machine translation.
Vol., 2015.
- 📄 **M. Luong, H. Pham, C. D. Manning.**
Effective approaches to attention-based neural machine translation.
In Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing, pp. 1412—1421, Lisbon, Portugal, 2015.
- 📄 **H. Mi, Z. Wang, A. Ittycheriah.**
Supervised attentions for neural machine translation.
arXiv preprint arXiv:1608.00112, Vol., 2016.
- 📄 **T. Mikolov, M. Karafiát, L. Burget, J. Cernocký, S. Khudanpur.**
Recurrent neural network based language model.
In Interspeech, Vol. 2, 3, 2010.

- 📄 **F. J. Och.**
Minimum error rate training in statistical machine translation.
In Proceedings of the 41st Annual Meeting on Association for Computational Linguistics - Volume 1, ACL '03, pp. 160–167, Stroudsburg, PA, USA, 2003. Association for Computational Linguistics.
- 📄 **H. Schwenk, M. R. Costa-Jussa et al.**
Smooth bilingual n-gram translation.
In EMNLP. Citeseer, 2007.
- 📄 **H. Schwenk.**
Continuous space language models.
Computer Speech & Language, Vol. 21, No. 3, pp. 492–518, 2007.
- 📄 **R. Sennrich, B. Haddow, A. Birch.**
Improving neural machine translation models with monolingual data.
Association for Computational Linguistics (ACL), Vol., pp. 86–96, 2016.

- 📄 **R. Sennrich, B. Haddow, A. Birch.**
Neural machine translation of rare words with subword units.
Association for Computational Linguistics (ACL), Vol., pp. 1715–1725, 2016.
- 📄 **S. Shen, Y. Cheng, Z. He, W. He, H. Wu, M. Sun, Y. Liu.**
Minimum risk training for neural machine translation.
Association for Computational Linguistics (ACL), Vol., 2016.
- 📄 **S. Sukhbaatar, J. Weston, R. Fergus et al.**
End-to-end memory networks.
In Advances in neural information processing systems, pp. 2440–2448, 2015.
- 📄 **M. Sundermeyer, R. Schlüter, H. Ney.**
Lstm neural networks for language modeling.
In Interspeech, pp. 194–197, 2012.

-  **I. Sutskever, O. Vinyals, Q. V. V. Le.**
Sequence to sequence learning with neural networks.
In Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, K. Weinberger, editors, *Advances in Neural Information Processing Systems 27*, pp. 3104–3112. Curran Associates, Inc., Monteval, Canada, 2014.
-  **Z. Tu, Z. Lu, Y. Liu, X. Liu, H. Li.**
Coverage-based neural machine translation.
CoRR, Vol. abs/1601.04811, 2016.
-  **Z. Tu, Z. Lu, Y. Liu, X. Liu, H. Li.**
Modeling coverage for neural machine translation.
In *54th Annual Meeting of the Association for Computational Linguistics*, 2016.
-  **A. Waibel, T. Hanazawa, G. Hinton, K. Shikano, K. Lang.**
Phoneme recognition: neural networks vs. hidden markov models vs. hidden markov models.
In *Acoustics, Speech, and Signal Processing, 1988. ICASSP-88., 1988 International Conference on*, pp. 107–110. IEEE, 1988.

- 📄 **Y. Wu, M. Schuster, Z. Chen, Q. V. Le, M. Norouzi, W. Macherey, M. Krikun, Y. Cao, Q. Gao, Klaus et al.**
Google's neural machine translation system: Bridging the gap between human and machine translation.
CoRR, Vol. abs/1609.08144, 2016.
- 📄 **K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. Zemel, Y. Bengio.**
Show, attend and tell: Neural image caption generation with visual attention.
In International Conference on Machine Learning, 2015.
- 📄 **L. Yao, A. Torabi, K. Cho, N. Ballas, C. Pal, H. Larochelle, A. Courville.**
Describing videos by exploiting temporal structure.
In Proceedings of the IEEE International Conference on Computer Vision, pp. 4507–4515, 2015.